1. $\lim_{n\to\infty}\left\{\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)\left(2^{\frac{1}{2}}-2^{\frac{1}{5}}\right)\ldots\left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)\right\}$ is equal to

[2023 (06 Apr Shift 2)]

(1) 1
(2) 0
(3) $\sqrt{2}$
(4) $\frac{1}{\sqrt{2}}$

2. $\lim_{x\to 0}\left(\left(\frac{1-\cos^2(3x)}{\cos^3(4x)}\right)\left(\frac{\sin^3(4x)}{(\log_e(2x+1))^5}\right)\right)$ is equal to

[2023 (08 Apr Shift 1)]

(1) 15
(2) 9
(3) 18
(4) 24

3. If $\alpha>\beta>0$ are the roots of the equation $ax^2+bx+1=0$, and $\lim_{x\to\frac{1}{\alpha}}\left(\frac{1-\cos(x^2+bx+a)}{2(1-\alpha x)^2}\right)^{\frac{1}{2}}=\frac{1}{k}\left(\frac{1}{\beta}-\frac{1}{\alpha}\right)$, then k is equal to

[2023 (08 Apr Shift 2)]

(1) 2β
(2) α
(3) 2α
(4) β

4. Among
$(S1)$: $\lim_{n\to\infty}\frac{1}{n^2}\left(2+4+6+\ldots+2n\right)=1$
$(S2)$: $\lim_{n\to\infty}\frac{1}{n^{16}}\left(1^{15}+2^{15}+3^{15}+\ldots+n^{15}\right)=\frac{1}{16}$

[2023 (13 Apr Shift 1)]

(1) Both $(S1)$ and $(S2)$ are true
(2) Only $(S1)$ is true
(3) Both $(S1)$ and $(S2)$ are false
(4) Only $(S2)$ is true

5. If $\lim_{x\to 0}\frac{e^{ax}-\cos(bx)-\frac{cxe^{-cx}}{2}}{1-\cos(2x)}=17$, then $5a^2+b^2$ is equal to

[2023 (13 Apr Shift 2)]

(1) 64
(2) 72
(3) 68
(4) 76

ANSWER KEYS

1. (2) **2.** (3) **3.** (3) **4.** (1) **5.** (3)

1. (2)

Let,
$L=\lim_{n\to\infty}\left\{\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)\left(2^{\frac{1}{2}}-2^{\frac{1}{5}}\right)\ldots\left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)\right\}$
Now,
$2^{\frac{1}{2}}-2^{\frac{1}{3}}<1$
$2^{\frac{1}{2}}-2^{\frac{1}{5}}<1$

$2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}<1\quad\forall n\in N$
And $\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)^n<\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)\left(2^{\frac{1}{2}}-2^{\frac{1}{5}}\right)\ldots\left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)<\left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)^n$
$\Rightarrow\lim_{n\to\infty}\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)^n<\lim_{n\to\infty}\left\{\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)\left(2^{\frac{1}{2}}-2^{\frac{1}{5}}\right)\ldots\left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)\right\}<\lim_{n\to\infty}\left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)^n$
$\Rightarrow\lim_{n\to\infty}\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)^n<L<\lim_{n\to\infty}\left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)^n$
And $\lim_{n\to\infty}\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)^n=0$ & $\lim_{n\to\infty}\left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)^n=0$
$\left\{\text{as } 2^{\frac{1}{2}}-2^{\frac{1}{3}}<1\ \&\ 2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}<1\right\}$
Hence, $\lim_{n\to\infty}\left\{\left(2^{\frac{1}{2}}-2^{\frac{1}{3}}\right)\left(2^{\frac{1}{2}}-2^{\frac{1}{5}}\right)\ldots\left(2^{\frac{1}{2}}-2^{\frac{1}{2n+1}}\right)\right\}=0$

2. (3)

Given,
$\lim_{x\to0}\left(\left(\frac{1-\cos^2(3x)}{\cos^3(4x)}\right)\left(\frac{\sin^3(4x)}{(\log_e(2x+1))^5}\right)\right)$
Now we now that,
$\lim_{x\to0}\frac{\sin x}{x}=1,\ \lim_{x\to0}\frac{1-\cos x}{x^2}=\frac{1}{2}\ \&\ \lim_{x\to0}\frac{\log(1+x)}{x}=1$
Now using the above formula we get,
$\lim_{x\to0}\left(\left(\frac{1-\cos^2 3x}{\cos^3 4x}\right)\left(\frac{\sin^3 4x}{(\log_e(2x+1))^5}\right)\right)$
$=\lim_{x\to0}\frac{(1-\cos 3x)(1+\cos 3x)9x^2}{(\cos^3 4x)9x^2}\frac{(\sin 4x)^3(64x)^3(2x)^5}{(64x^3)(\log_e(2x+1))^5(2x)^5}$
$=\lim_{x\to0}\frac{\left(\frac{1-\cos 3x}{9x^2}\right)(1+\cos 3x)}{(\cos^3 4x)}\frac{\left(\frac{\sin 4x}{4x}\right)^3}{\left(\frac{\log_e(2x+1)}{2x}\right)^5}\times\frac{9\times64}{32}$
$=\frac{\left(\frac{1}{2}\right)\times(2)}{(1)}\frac{(1)^3}{(1)^5}\times\frac{9\times64}{32}$
$=\frac{9\times2}{1}=18$

3. (3)

Since, α and β are the roots of the equation $ax^2+bx+1=0$, therefore $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ would be the roots of $x^2+bx+a=0$.

Let

$L=\lim_{x\to\frac{1}{\alpha}}\left[\frac{1-\cos\left(x^2+bx+a\right)}{2\left(1-\alpha x\right)^2}\right]^{\frac{1}{2}}\to\frac{0}{0}$

$\Rightarrow L=\lim_{x\to\frac{1}{\alpha}}\left[\frac{2\sin^2\left(\frac{x^2+bx+a}{2}\right)}{2\left(1-\alpha x\right)^2}\right]^{\frac{1}{2}}$

$\Rightarrow L=\lim_{x\to\frac{1}{\alpha}}\left[\frac{\sin^2\left(\frac{1}{2}\left(x-\frac{1}{\alpha}\right)\left(x-\frac{1}{\beta}\right)\right)}{\alpha^2\left(x-\frac{1}{\alpha}\right)^2}\right]^{\frac{1}{2}}$

$\Rightarrow L=\lim_{x\to\frac{1}{\alpha}}\left[\frac{\left(x-\frac{1}{\beta}\right)^2}{4\alpha^2}\times\frac{\left\{\sin\left(\frac{1}{2}\left(x-\frac{1}{\alpha}\right)\left(x-\frac{1}{\beta}\right)\right)\right\}^2}{\left(\frac{1}{2}\right)^2\left(x-\frac{1}{\alpha}\right)^2\left(x-\frac{1}{\beta}\right)^2}\right]^{\frac{1}{2}}$

$\Rightarrow L=\frac{\left(\frac{1}{\alpha}-\frac{1}{\beta}\right)}{2\alpha}$

On comparing this value with the value given in the question we get, $k=2\alpha$.

Hence this is the correct option.

4. (1)

$S_1:\lim_{n\to\infty}\frac{1}{n^2}\left[2+4+6+\ldots+2n\right]$

$\lim_{n\to\infty}2\frac{n(n+1)}{2n^2}=1$

$S_2:\lim_{n\to\infty}\frac{1}{n^{16}}\left(1^{15}+2^{15}+3^{15}+\ldots+n^{15}\right)$

This is limit of sum.

We know that $\lim_{n\to\infty}\sum_{r=1}^{n}f\left(\frac{r}{n}\right)\frac{1}{n}=\int_0^1 f(x)\mathrm{d}\,x$

$\because\lim_{n\to\infty}\frac{\sum_{r=1}^{n}r^k}{n^{k+1}}=\lim_{n\to\infty}\frac{\sum_{r=1}^{n}\left(\frac{r}{n}\right)^k}{n}$

$=\int_0^1 x^k\,\mathrm{d}\,x=\frac{1}{k+1}$

Here $k=15$

$\therefore\lim_{n\to\infty}\frac{\sum_{r=1}^{n}r^{15}}{n^{16}}=\frac{1}{16}$

$\therefore$ Both $S1$ and $S2$ are correct

Hence this is the required option.

5. (3)

Given that

$\Rightarrow\lim_{x\to0}\frac{e^{ax}-\cos\left(bx\right)-\frac{cx}{2}e^{-cx}}{1-\cos 2x}=17$

We know that $e^{ax}=1+ax+\frac{(ax)^2}{2!}+\ldots$ and $\cos(bx)=1-\frac{(bx)^2}{2!}+\ldots$

$\Rightarrow\lim_{x\to0}\frac{\left(1+ax+\frac{(ax)^2}{2!}+\ldots\right)-\left(1-\frac{(bx)^2}{2!}+\ldots\right)-\frac{cx}{2}\left(1-(cx)+\frac{(cx)^2}{2!}-\ldots\right)}{\left(\frac{1-\cos 2x}{(2x)^2}\right)\times4x^2}=17$

$\Rightarrow\lim_{x\to0}\frac{\left(a-\frac{c}{2}\right)x+\left(\frac{a^2+b^2+c^2}{2}\right)x^2+\ldots}{\frac{1}{2}\times4x^2}=17$

Now for limit to exist, $a-\frac{c}{2}=0\Rightarrow c=2a$

$\Rightarrow\lim_{x\to0}\frac{\left(\frac{a^2+b^2+c^2}{2}\right)x^2+\ldots}{\frac{1}{2}\times4x^2}=17$

$\Rightarrow\frac{a^2+b^2+c^2}{4}=17$

$\Rightarrow a^2+b^2+4a^2=68$

$\Rightarrow 5a^2+b^2=68$

Hence this is the correct option.